

# Report Sheet

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## Phase-Locking of Local Oscillations

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### 1 Introduction:

Macroscopic oscillations ranging from infraslow to ultrafast, in different brain regions play a crucial role in routing information across cortical regions through cross-coupling mechanisms based on Communication through Coherence (CTC) theory.

In this project, we followed the work of Dumont and Gutkin (2019), focusing on Pyramidal-Interneuron Network Gamma (PING) oscillations within the 30–150 Hz frequency band. Coherent gamma oscillations have been reported in numerous brain regions across various species and are associated with a wide range of cognitive tasks. The Communication Through Coherence (CTC) theory proposes that these phase-locking states may regulate information flow. Specifically, there is a temporal window of excitability within the oscillatory cycle during which the excitatory population is more likely to respond to stimulation. The oscillatory firing rate modulates the network's excitability rhythmically; therefore, two oscillating neural groups communicate more efficiently when they maintain a coherent phase relationship.

Spike transmission between networks is not instantaneous; it may take tens to hundreds of milliseconds. Synaptic delays are included to account for the long-range distances separating the circuits. Consequently, oscillations become lagged. This synaptic delay plays a central role in producing a non-symmetric phase lag between macroscopic oscillations in two coupled networks, where one network begins to lead the other in phase. These varying phase lags are associated with different phase-locking modes. Such a collection of phase lags has been suggested to facilitate the control and selection of information flow through anatomical pathways.

The Macroscopic Phase Response Curve (mPRC) determines how the phase of a global oscillation responds to incoming perturbations. A short depolarizing current injected into the network induces a phase shift that depends not only on the perturbation strength but also on the phase at which the perturbation is applied. Furthermore, PING and ING rhythms exhibit distinct mPRC types, which are used to classify these rhythmic regimes. Finally, the mPRC serves as a fundamental tool to study differences in locking phases under periodic pulsatile stimuli in oscillating systems composed of different cell types.

First, we implemented a network of Quadratic Integrate-and-Fire (QIF) neurons and simulated a single E-I circuit. Second, we explored the effects of varying external inputs to the excitatory and inhibitory populations on the network's oscillation coherence and frequency. Third, we coupled two

E-I networks and estimated the phase lag as a function of the synaptic delay and synaptic strengths between the two coupled circuits. Finally, we reproduced the Phase Response Curve (PRC) by injecting small perturbations into the local network at different phases.

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## 2 Steps one and two: Implementing Network Dynamic Equations and Simulation E-I Network

Each cell is described by the quadratic integrate-and-fire (QIF) model:

$$\tau \frac{d}{dt} v_j = \eta_j + v_j^2 + I \quad (1)$$

To account for the network heterogeneity, the intrinsic parameter  $\eta$  is distributed randomly according to Lorentzian distribution:

$$L(\eta) = \frac{1}{\pi} \frac{\Delta}{((\eta - \bar{\eta})^2 + \Delta^2)} \quad (2)$$

This offers a wide range of intrinsic excitability when  $\Delta$  is large.

$$\int_{-\infty}^0 L(\eta) d\eta = \frac{1}{\pi} \left( \frac{\pi}{2} - \arctan \left( \frac{\bar{\eta}}{\Delta} \right) \right) \quad (3)$$

For  $\Delta > 0$ , this integral is not zero, meaning not all of them are intrinsically oscillating.

The total synaptic current,  $I(t)$  is assumed to be the sum of an external input and the synaptic inputs  $s_e$  and  $s_i$ :

$$I_e = I_e^{ext} + \tau_e s_{ee} - \tau_e s_{ei} \quad (4)$$

$$I_i = I_i^{ext} + \tau_i s_{ie} - \tau_i s_{ii} \quad (5)$$

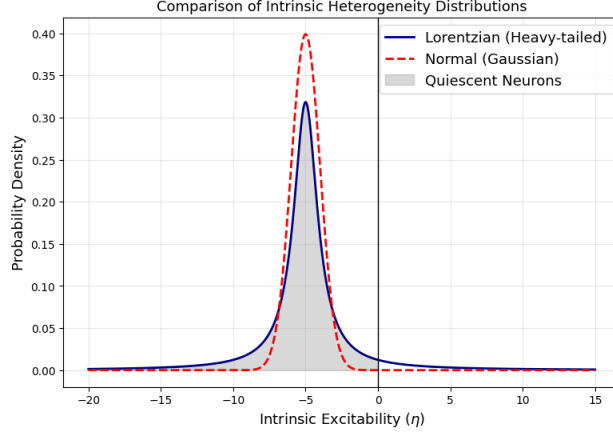
The dynamics of synaptic current,  $s(t)$  for E-cells is described as:

$$\begin{cases} \tau_s \frac{d}{dt} s_{ee} = -s_{ee} + J_{ee} r_e \\ \tau_s \frac{d}{dt} s_{ei} = -s_{ei} + J_{ei} r_i \end{cases} \quad (6)$$

respectively for I-cells:

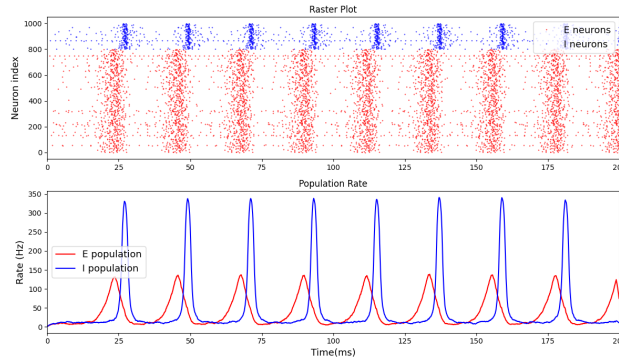
$$\begin{cases} \tau_s \frac{d}{dt} s_{ii} = -s_{ii} + J_{ii} r_i \\ \tau_s \frac{d}{dt} s_{ie} = -s_{ie} + J_{ie} r_e \end{cases} \quad (7)$$

Having established the microscopic dynamics of individual neurons and their synaptic interactions, we proceed to simulate the collective behavior of the E-I network.



This distribution illustrates that with a mean excitability of  $\bar{\eta} = -5$ , the majority of neurons are not intrinsically oscillatory. Consequently, the emergence of a global oscillation in the network is driven primarily by synaptic currents rather than individual neural dynamics.

The network is constituted of one excitatory (E) and one inhibitory (I) population. In this configuration, a constant external input ( $I_e^{ext} = 10$ ) is provided to the E-cells, while the I-cells receive no direct external drive ( $I_i^{ext} = 0$ ). This input causes the E-cells to fire first; their spikes then provide a recruitment signal that excites the I-cells. Once the I-cells spike, they provide feedback inhibition that briefly silences the E-population. As this inhibition decays, the E-cells reach their firing threshold again, restarting the cycle. This rhythmic interplay between excitation and delayed inhibition generates the PING oscillation observed in our results.



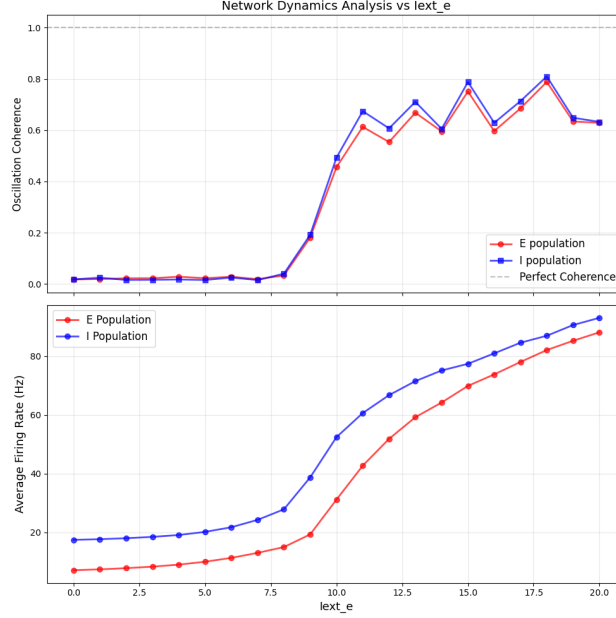
### 3 Step three : The effects of varying relevant parameters

Now we want to explore the effects of varying relevant parameters, like the external input to E-cells and I-cells.

In this section we defined four functions to visualise the change of oscillation coherence and firing rate of populaitions:

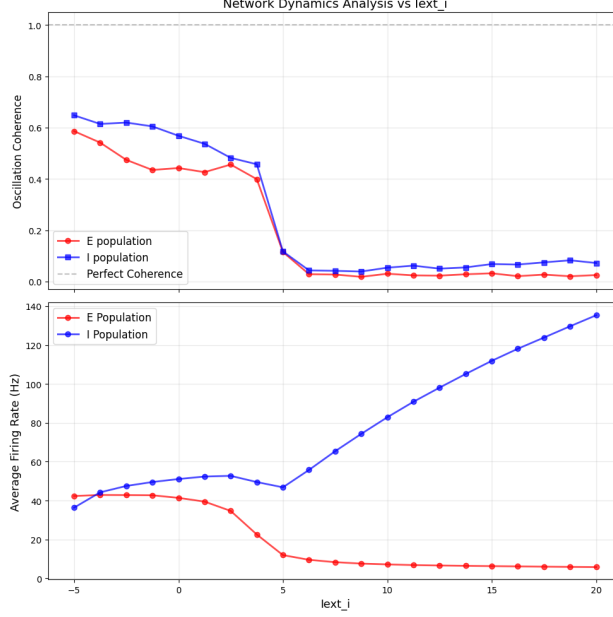
1. *1st Function*, “compute\_power\_spectrum” calculates the Power Spectrum of the network activity. It returns ‘freqs’ (Hz) and ‘power’ (Hz<sup>2</sup>) to identify dominant oscillation speeds.

2. *2nd function*, “compute\_coherence” Returns the oscillation coherence and the peak frequency. The oscillation coherence is quantified as the ratio of the peak power(in Gamma band) to the mean power of the spectrum  $\sqrt{\frac{\text{peak power}}{\text{mean power}}}$ . This indicates that how much of the total firing energy is concentrated into highest frequency in Gamma band versus how much is wasted as stochastic, asynchronous noise. Coherence quantifies the strength of the rhythm, particularly as we vary parameters like external input, ensuring that a high coherence value truly reflects a concentrated temporal alignment of spikes across the population.
3. *3th function*, “run\_ping\_network” simulates a spiking neural network consisting of Excitatory (E) and Inhibitory (I) populations based on the Quadratic Integrate-and-Fire (QIF) model. It utilizes PopulationRateMonitor to record the collective firing rate of each group.
4. *4th function*, “parameter\_sweep” automates the process of investigating how the network’s dynamics change under different conditions. It iterates through a defined range of values for external input while keeping others constant. Fixed values for external input currents to exticatory and inhibitory populations are 10 and 3, respectively. It returns the coherence of both populations’ oscillation and their average firing rate.



We fix the external input to inhibitory population at 3, and change the excitatory one from 0 to 20, to check its effects on network activity:

Between  $I_e^{ext} = 0$  and 8, the coherence is near zero. This indicates an Asynchronous State where neurons fire randomly without coordination. Near  $I_e^{ext} = 9$ , we observe a sharp onset of synchrony, indicating a Hopf Bifurcation where the system transitions into a limit cycle (PING oscillation). And the coherence stabilizes between 0.6 and 0.8. This shows a robust, stable oscillation. The red (E) and blue (I) lines are almost perfectly identical. This confirms that the inhibitory neurons are tightly “locked” to the excitatory rhythm. The transition to limit cycle is accompanied by a significant increase in the firing rates of both populations, with the inhibitory neurons consistently maintaining a higher rate to enforce the rhythmic pacing of the network and creating a temporal and rhythmic excitability window that define the Gamma rhythm.

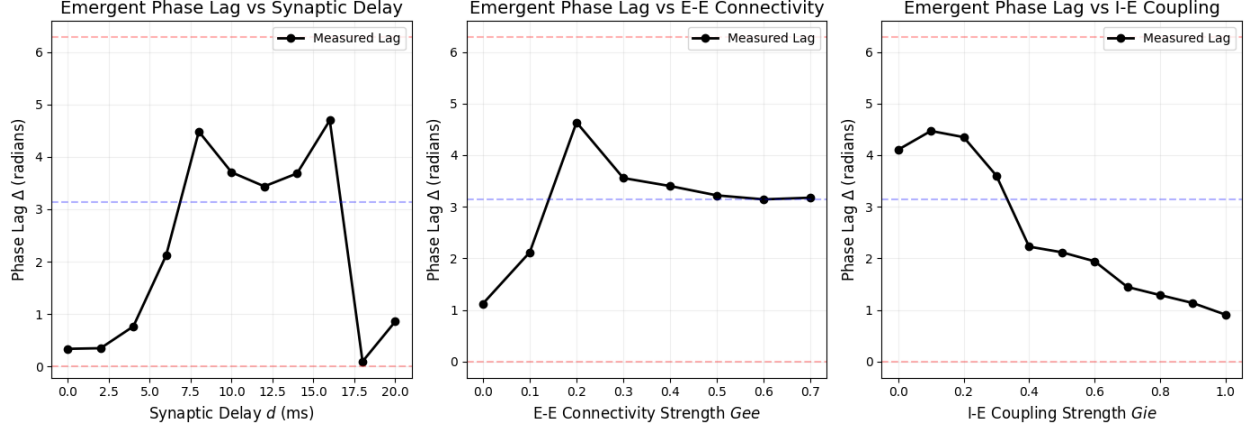


We fix the external input to excitatory population at 10, and change the inhibitory external input from  $-5$  to  $20$ , to check its effects on network activity:

In contrast to the excitatory sweep, the network exhibits high coherence only when the inhibitory external drive is low ( $-5 \leq I_i^{ext} \leq 4$ ). In this regime, the inhibitory population is primarily recruited by the excitatory neurons, allowing for a functional PING mechanism where E-cells lead and I-cells pace the rhythm. But, the stronger external drive to the inhibitory population ( $I_i^{ext} > 5$ ) disrupts the PING mechanism where the stable limit cycle collapses. Because there is no coherence in the oscillation of the network. The excessive drive to the inhibitory population close the rhythmic excitability window. The average firing rate plot shows that by increasing input current to inhibitory population, the firing rate in excitatory population decreases but not completely silenced—continuing to fire at a low rate—their activity becomes desynchronized. The loss of E-cells coordination leads to a paradoxical effect on the I-population: Although the inhibitory firing rate increases, their spikes lose all coherence. This occurs because the inhibitory cells are no longer receiving the structured, periodic pulses from the E-cells that previously facilitated their synchronization.

## 4 Step 4: Coupled PING Networks (mPRC)

In this section we think of a good way of estimating the lag in coupled two E-I networks. Then we explore how the lag varies as a function of delay and synaptic strength, cross-E $\rightarrow$ E and cross-E $\rightarrow$ I connection.

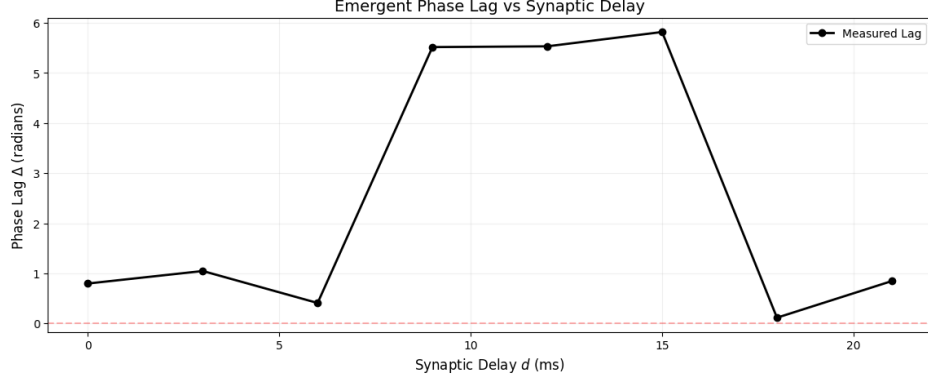


**Initial Simulations:** These graphs were attempts to reproduce the bifurcation diagrams shown in figure 6 of the Dumont and Gutkin paper.

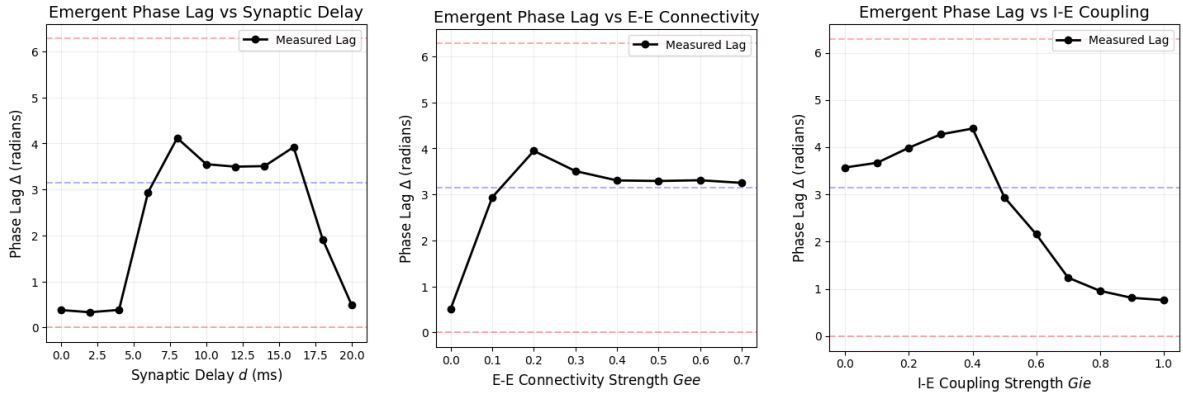
For synaptic delay, the plot suggests that for short and long delays the network produces in-phase oscillations, while for an intermediate range the network produces anti-phase oscillations. However if we look at the raster plots associated with specific delay values it is obvious that for short and long delays we are in a leader/follower regime biased towards in-phase oscillations. For  $d=12$  it might be that we are approaching a value near some bifurcation that would result in anti-phase oscillations. Alternatively it could be that this value simply results in a leader/follower regime with high period such that the simulation could not accurately estimate the phase lag.

Our simulations differ from Dumont and Gutkin by the number of neurons used and a different spiking values. It could be that these simulations are not using enough neurons which results in slower convergence. We could not feasibly produce these graphs with larger populations. Alternatively, changing the reset/threshold values without changing all the other related parameters might have also changed where we are in parameter space. As a result, we are not close enough to a limit cycle regime (hop bifurcation?) that produces the effects as seen in Dumont and Gutkin.

For coupling strength parameters  $G_{ie}$  and  $G_{ee}$ , we can see some similar effects as described by Dumont and Gutkin. Increasing E-E leads to a bifurcation resulting in anti-phase oscillations. We find that before this point the phase is neither strongly in or out of phase. Here there is leader follower dynamics. For I-E coupling, the dynamics go from leader follower to in-phase oscillations. Both of these effects are clearly continuous are reflective of some sort of bifurcation. For coupling values near 0, Dumont and Gutkin report clear in-phase/anti-phase oscillations (respectively), however our simulations do not produce these.



The above plot was made to test whether the number of neurons were effecting the results. This graph is nearly identical to the previous phase lag vs delay plot however it uses 5000 neurons (as opposed to 2000) and different values  $d$  are sampled. Since  $d=12$  was not sampled here, a simulation was run independently and the dip toward  $\pi$  still occurs. The raster plots still show that we are in a leader follower regime and not in oscillating in/out of phase.



**Refined Results:** The previous phase lag vs control parameter plots were not correctly scaled because the reset/threshold parameters were reduced by a factor of 10. The three plots above used the parameters as described in the paper with the brian2 defaultclock set to 0.01ms instead of 0.1ms. This produces results similar to what Dumont and Gutkin reported. The issues with the raster plots as described previously do not occur.

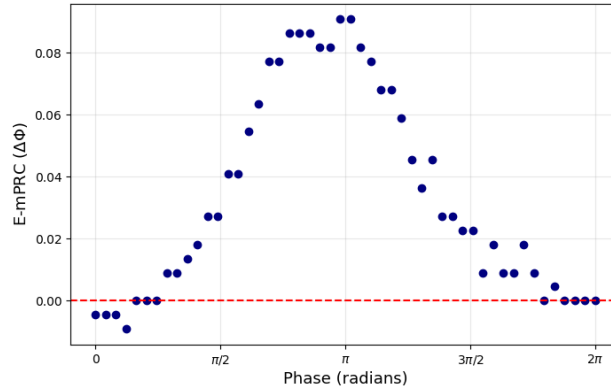
For synaptic delay, the plot still suggests that for short and long delays the network is in an in-phase oscillations regime. For a short delay range between 6-8ms there we are in a non-symmetric regime. Between 10ms-14ms the network produces anti-phase oscillations.

These simulations still differ from the paper because only 2000 neurons are used. It could be that these simulations are not using enough neurons which results in slower convergence making the bifurcation more difficult to detect. We could not feasibly produce these graphs with larger populations.

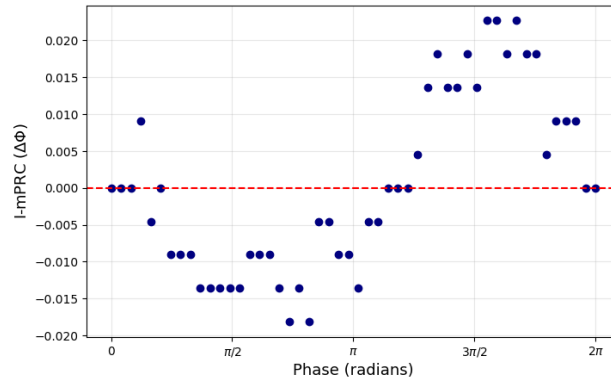
For coupling strength parameters  $G_{IE}$  and  $G_{EE}$  the results are quite similar to what we saw previously.

## 5 Step 5: Microscopic Phase Resetting Curve (mPRC)

We want to try to reproduce the phase response curve in the local network. In this regard, We utilize the `run_ping_network` function to generate a baseline reference signal. After discarding the initial transient activity, we calculate the mean oscillation period ( $T$ ) by averaging the intervals between successive peaks of the population firing rate. We then implement the `run_ping_pulse` function to simulate the network's response to external perturbations. By injecting a brief excitatory pulse at various phases ( $\phi$ ) within the oscillation cycle—targeting either the Excitatory (E) or Inhibitory (I) populations—we measure the resulting phase shift. This allows us to observe how the timing of a perturbation can either advance or delay the subsequent oscillation peaks.



When an excitatory pulse is injected into the E-cells, the network rhythm typically exhibits a monophasic response. The pulse provides additional excitation directly to the excitatory population. This leads exclusively to a phase advance, meaning the next cycle always occurs earlier regardless of the injection phase.



In our code, the period is calculated based on the successive peaks of the inhibitory population's firing rate. When an excitatory pulse is injected into the I-cells, we observe a biphasic response:

**First Half (Phase Delay):** If the pulse occurs during the first half of the cycle—when I-cells are already firing or have just finished—it provides extra excitation to the inhibitory population. This strengthens or prolongs the inhibition felt by the E-cells. As a result, the E-cells take longer to recover and reach their threshold, which delays the next cycle.



Second Half (Phase Advance): If the pulse is injected during the second half of the cycle—approaching the time when E-cells would normally trigger the next round of I-cells—it causes the I-cells to fire sooner than expected. By ‘forcing’ the inhibitory peak to occur earlier, the current cycle is completed faster, allowing the next cycle to start sooner (advance).

It is important to note that the amplitudes of PRCs show us the network is more sensitive to perturbations onto excitatory cells than inhibitory cells.

## References

- [1] Dumont, G., & Gutkin, B. (2019). *Macroscopic phase resetting-curves determine oscillatory coherence and signal transfer in inter-coupled neural circuits*. PLoS computational biology, 15(5), e1007019.